CLOUD COMPUTING

CAC-1

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Application of Cloud computing in Geoscience

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ABSTRACT

Cloud computing services and their importance in geoscience.

Scalability and connectivity are important aspects of cloud computing for geoscience.

GEOSS is an example of a successfully established and running example of cloud computing in the field of geoscience

The GEOSS clearinghouse was installed, maintained, and tested on the Amazon Elastic Cloud Computing (EC2) platform to evaluate the usage of cloud computing for geosciences applications.

EC2 cloud computing platform benefits geospatial applications in the following ways: a) scalability, b) dependability, and c) minimizing duplication of effort among geosciences communities.

Test of concurrent user requests for big data queries demonstrates the need for different applications to be justified and optimized before being deployed into the EC2 platform for a better balance of cost and performance.

INTRODUCTION

Computing platforms and paradigms advance due to technological advances like multi-core CPUs and networked computing environments. In the last few decades, computer paradigms have changed to become cloud computing.

How to set up an optimum computing platform for geospatial analysis and other classic computing paradigms is a significant problem for high-performance and parallel computing.

Great computational power, memory, and quick network connections are always preferred to manage the jobs. Customers must pay for the resources they use in cloud computing, especially for commercial cloud services billed or charged using a consumption-based approach.

So, when setting up the cloud computing infrastructure to enable geospatial applications, pricing is a crucial issue to consider.

To satisfy diverse computing needs, Amazon EC2 offers a variety of instance kinds, with each instance offering a varying level of dedicated compute resources, such as CPU power, memory, and disc space. To satisfy computational needs.

CLOUD SERVICES

The types of services that cloud computing now offer include are categorized as Software as a Service (SaaS), Platform as a Service (PaaS), and Infrastructure as a Service (IaaS).

The computer is provided through IaaS. Infrastructure, such as a network, storage, virtual server grid, cluster, and operating system software, are standardized network services.

A computer platform is provided as a service through PaaS. It includes a software layer and offers it as a service that can be accessed to Create services at a higher level. Users can run existing apps or create new ones without worrying about maintaining the operating system, server infrastructure, load balancing, or computational power on such a platform.

SaaS offers all the features found in complex, locally installed traditional apps, making it the most well-known and popular kind of cloud computing. These capabilities to end-users distinguish these local applications from SaaS.

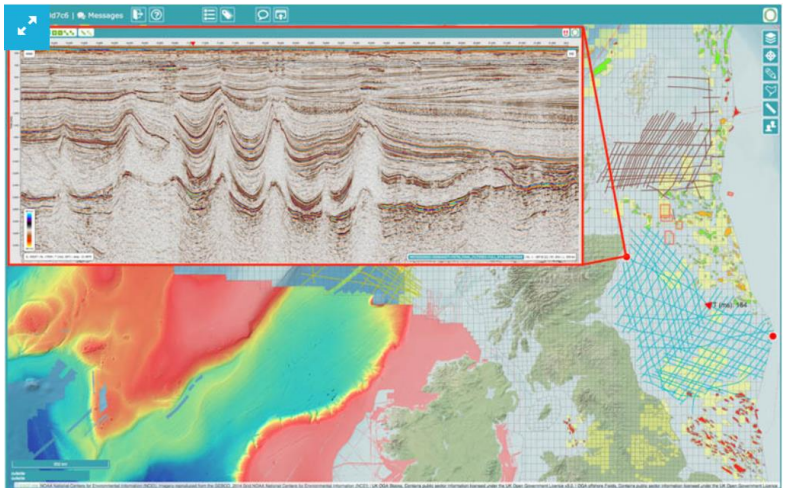


FIGURE-1:

Example of viewing subsurface data through a web browser.

For a geoscientist, the key opportunities are the scalability and connectivity of the cloud, as well as its near-limitless compute power.

SCALABILITY OF THE CLOUD

Depending on the user's requirement, the cloud can scale up or down. This implies that the cloud can quickly scale up and test the same property across the whole seismic repository and test it on a single seismic line. Operations can execute in parallel even if you have 100 times as much data. Thus, retrieving the results won't take 100 times as long. Large-scale portfolio reviews and data filtering could benefit significantly from this.

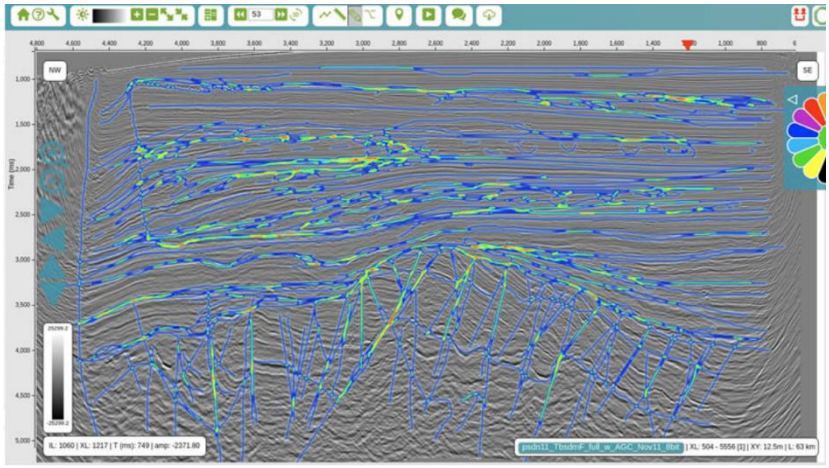


FIGURE-2:

Example of an interpretation heatmap generated from crowdsourced interpretation.

CONNECTIVITY

The choice to safely exchange data with third parties is provided by cloud connectivity. By sharing knowledge and creating fresh insights into the data, the engagement with joint ventures and academic alliances can be strengthened. Internally, the geoscience workflow can be improved by using the cloud's connectivity. To provide crowd-sourced interpretations and alternative perspectives on the same data, data can be shared internationally and evaluated in a matter of seconds. This enables a wide range of alternative concepts and provides a better knowledge of a given asset's challenges and full potential.

Once the data is on the cloud, It quickly connects to the most recent advancements in various industries, which will provide a distinct competitive advantage. Both deep learning algorithms and data scientists can transform data to be suitable for use. The current geoscience workflows can be re-interpreted, fresh insights may be acquired, and new opportunities can be curated by fusing knowledge and domain expertise with the latest technologies.

AMAZON EC2

Amazon EC2 is a critical component of Amazon's cloud services that enable users to deploy scalable resources as needed.

A typical IaaS cloud service is Amazon EC2. EC2 allows users to boot an Amazon Machine Image (AMI) to construct a virtual machine, or what Amazon refers to as an "instance," which is based on Xen.

A bootable virtual machine (VM) root image, or AMI, contains several operating systems and any software required to build a VM.

To satisfy computational needs, EC2 now provides a variety of instance kinds, each of which delivers a predictable amount of dedicated compute capability (CPU power, memory, disc, etc.).

These EC2 instances are divided into three categories by Amazon: Standard, High-Memory, and High-CPU Linux illustrations.

Each category is best suited for a specific type of application.

DEPLOYING APPLICATIONS ONTO THE CLOUD

Users can ask to launch an instance based on a specific AMI through the AWS (Amazon Web Service) Management Console or Amazon EC2 AMI Tools.

Upon approval of the request, a VM is set up. The two storage services offered by Amazon include Simple Storage Service and EBS (Elastic Block Store). Both storage forms can serve AMI volume storage and the Amazon EC2 instance's virtual storage device. An AMI needs to be ready before you can deploy applications on Amazon.

There are two typical methods in Linux for setting up an AMI:

1) The most straightforward approach is to start with a customizing public AMI and customize it to the most specific needs.

2) Another method is to build a new installation on a standalone computer or an empty file system mounted by loopback. This applies to both Amazon EBS-backed and Amazon S3-backed AMIs.

This only applies to AMIs backed by Amazon S3 and requires starting from scratch with an operating system installation.

DEPLOYING GEOSS CLEARINGHOUSE ONTO AMAZON EC2

GEOSS Clearinghouse- A typical search tool for the Intergovernmental Group on Earth Observation is the GEOSS (Global Earth Observing Systems) Clearinghouse (GEO).
EO data, services, and related resources can be found and accessible through the harvesting or distributed searching registered metadata catalogs.
It is now being deployed to the Amazon EC2 cloud platform while bested on a CISC server.

Deployment-



FIGURE-3:

The process of deploying GEOSS Clearinghouse onto Amazon EC2

STEP 1: The GEOSS Clearinghouse AMI is created, which involves tweaking an already-existing public AMI. To start an Amazon EC2 instance, a public AMI with CentOS 5.4 as the operating system is chosen. Users can interact with the EC2 instance once it has been loaded with the chosen AMI and has started booting up.

STEP 2: After allowing network access by opening port 22, which enables SSH access, the next step is to log into the instance to gain complete root access via the remote accessing method SSH (Secure Shell). The AWS Management Console or command-line tools from Amazon can be used to configure this.

STEP 3: After login in, the user is free to explore and experiment with the system as they choose, for example, by setting up FTP and dropping a web application. In the case of our organization, the servlet container tomcat and the database software Postgresql with PostGIS to enable spatial datasets, which hold the data of GEOSS Clearinghouse, should be deployed.

The EBS volume can be cloned and used to restore the database on a second EC2 instance. Both command-line tools and the AWS Management Console can be used to create and attach the EBS volume to the GEOSS Clearinghouse instance. After successfully launching the Amazon EC2 instance, users are ready to begin exploring the endless EC2 possibilities.

Amazon SQS provides a good solution for the GEOSS Clearinghouse to automatically scale up and down the instances. Scalable EC2 services would facilitate geospatial applications, elastically satisfying computing requirements. Used with Amazon EC2, as well as Amazon S3, Amazon SQS can make applications more flexible and scalable.

The stability of geospatial applications is increased via Amazon EC2. Due to the service's usage of Amazon's tried-and-true network and datacenter infrastructure, Amazon EC2 provides a highly dependable environment for the GEOSS Clearinghouse.

All Amazon EC2 regions, including US Standard, EU (Ireland), US West (Northern California), and the Asia Pacific, are guaranteed by the Amazon EC2 Service Level Agreement (SLA) to have 99.95 percent availability (Singapore). The GEOSS Clearinghouse data are consequently safeguarded from instance termination or failure because the EBS data volume can be connected to another instance.

AMAZON EC2 PLATFORM TEST

The effectiveness of various software categories is tested using GetCapabilities requests from multiple amounts of concurrent user queries. Regarding supporting the GEOSS, Amazon EC2 instances Clearinghouse GetCapabilities is a crucial CSW (catalog search tool) GEOSS clearinghouse service online request.

Once the GetCapabilities request is sent to the GEOSS clearinghouse server-side request, it will first read a file containing an XML template and fill it out by reading the local configuration files and populating the template with data Lucene to search a database; the filled-out XML template will be returned to the customer.

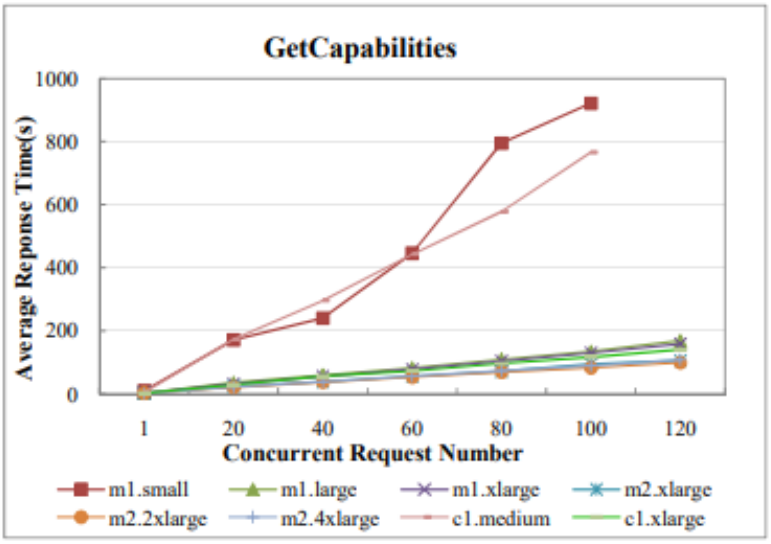


Figure- 4: GetCapabilities Performance Comparison by Amazon's eight types of instances

Figure 4 shows the performance of the eight Amazon EC2 instances by different concurrent user requests. Performance is illustrated with eight-line graphs with six lines close together, indicating the superior performance of the Large instance (m1.large). These six instances have faster response times than the other two, including small and high CPU media instances (c1.medium). Extra-large instance (m1.2xlarge), High memory extra-large (m2.4xlarge), double extra-large (m2.2xlarge), Quadruple extra-large instances (m2.4xlarge), and High-CPU extra-large instance (c1.xlarge). These six instances show very little difference in performance even though they have different CPU cores and memory numbers.

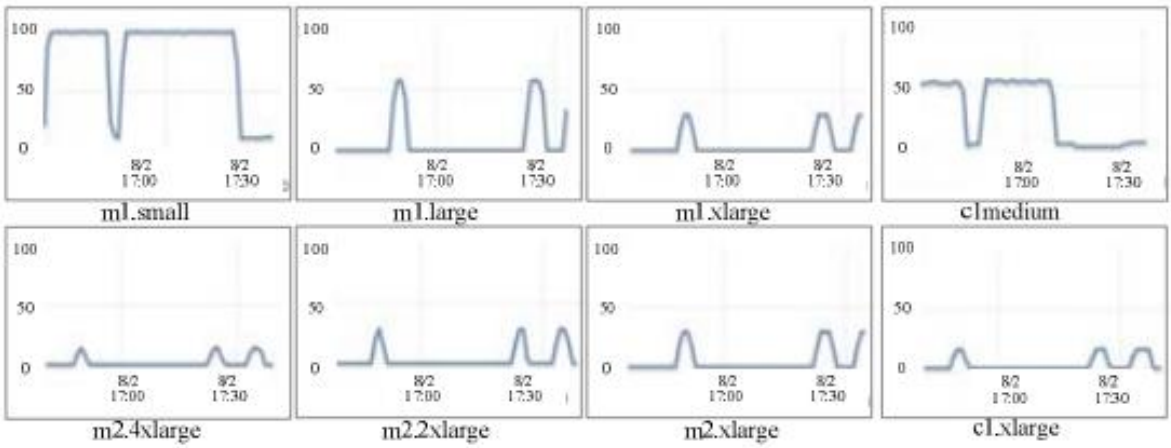


Figure- 5: Average CPU Utilization of Amazon's 8 types of instances

The six cases have higher performance even if there is not much of a variation in their performance contrasted with the previous two cases (m1.small, c1.small).

Taking Standard Linux into account cost and performance to balance the price and size, a large prominent (m1.large), given that it only costs about 0.34 and is relatively high quite effective.

CONCLUSION

For Geosciences, spatial cloud computing is the key to constructing such a spatial computing environment; a key issue is how to incorporate the underlying resources through Geospatial middleware.

Implementing a Geospatial middleware for cloud computing is more complex than other computing paradigms.

In addition, the EC2 cloud computing platform can facilitate geospatial applications in the aspects of a) scalability, b) reliability, and c) reducing duplicated efforts among Geosciences communities.

Through the deployment and maintenance of the GEOSS Clearinghouse on the EC2 platform, we demonstrated how to utilize cloud computing to support Geosciences applications. Other issues related to cloud computing are data and personal privacy, unpredictable performance, data transfer bottlenecks, bugs in large-scale distributed systems, software licensing, and quick Scaling.

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